

Mukhlis Amien

Lecturer in Informatics | Researcher in ML, NLP & Computational Data Science

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RESEARCH INTERESTS

Machine Learning · Natural Language Processing · Knowledge Discovery · Deep Learning for Bioinformatics · Computational Spatial Analysis · Low-Resource NLP · Bayesian Modelling

EDUCATION

M.Sc. in Computer Science (Magister Ilmu Komputer)

Universitas Indonesia, Jakarta | 2016 | GPA: 3.42 / 4.00

Thesis: Feature Selection Method Using Multi-Step Weight-Based Rank Using Deep Learning to Search Biomarker in Microarray Data

B.Sc. in Informatics Engineering (Sarjana Teknik Informatika)

STIKI Malang | 2001 | GPA: 2.92 / 4.00

ACADEMIC POSITION

Lecturer (Dosen Tetap) | Universitas Bhinneka Nusantara (UBHINUS) | 2010 – Present

- Courses: Machine Learning, Computer Vision, AI, NLP, Mobile & Web Programming
- Academic rank: Asisten Ahli (Assistant Lecturer), since December 2025
- Supervisor of undergraduate final projects in ML/NLP
- Active researcher with ongoing international publication programme (4 papers under review, 3 in drafting, 2026)

Head of International Affairs | STIKI Malang | 2012 – 2014

- Coordinated international seminars and student/faculty exchange programmes

PREVIOUS PROFESSIONAL EXPERIENCE

Founder & Director | CV. Jaring Digital Solusi | 2004 – 2008

- IT consulting, HRMIS development, local government IT projects

System Analyst & Programmer | PT. Myoh Technology Tbk | 2001 – 2002

- Hotel and cooperative information systems

CURRENT RESEARCH PROJECT

VOLCARCH — Volcanic Taphonomic Bias in Indonesian Archaeological Records (2025–present)

Principal investigator. Applies ML, NLP, Bayesian simulation, and GIS to reconstruct pre-colonial settlement patterns in Java. Core challenge: ~99.94% of expected signals absent due to volcanic burial (4.4 ± 1.2 mm/yr, $r = 0.951$, calibrated across 3 volcanic systems).

- 175 tracked experiments across 6 evidence layers (E001–E175, ongoing)
- Monte Carlo: 100,000 runs; population estimate 3.3M (7/7 calibration points)
- NLP pipeline: 22,000+ colonial Dutch (VOC) texts; rule-based IE pipeline v0.1 operational

- Spatial suitability model: AUC 0.768; 3.5x search efficiency gain over random survey
- 4 papers under review (EGQJSJ, JCAA, Antiquity, Oceanic Linguistics); 3 in drafting (Archipel, DHQ, ArchCalc)
- Preprint (Zenodo): DOI 10.5281/zenodo.19081502 | arXiv:2604.00023
- Public code + data: github.com/neimasilk/volcarch-repo
- FDR audit: 65/83 tests survive Benjamini-Hochberg (78.3%)

PUBLICATIONS

Under Review (2026)

- Amien, M. "Taphonomic framework and cascade model." E&G Quaternary Science Journal (egqsj-2026-3).
- Amien, M. & Gunawan, G.F. "GIS-based settlement suitability modelling." Journal of Computer Applications in Archaeology (#280).
- Amien, M. "Temporal Overlay Matrix for archaeological visibility." Antiquity (Project Gallery).
- Amien, M. & Gunawan, G.F. "Phonological fossils: ML detection of non-mainstream vocabulary." Oceanic Linguistics. arXiv:2604.00023.

In Drafting (2026)

- Amien, M. "Temples without villages: Settlement hierarchy gaps in 9th–10th century Java." Archipel (INALCO/EHESS).
- Amien, M. "Computational textual archaeology of the Old Javanese epigraphic corpus." Digital Humanities Quarterly.
- Amien, M. "Two Javas: Sacred vs. administrative space in the epigraphic record." Archeologia e Calcolatori (MS: 365).

Published / Preprint

- Gunawan, G.F. & Amien, M. (2026). "LLM benchmark on software engineering tasks." arXiv:2602.07079.
- Amien, M. & Gunawan, G.F. (2024). "BERT dan Bahasa Indonesia." ELANG Journal 1(2), 132–140.
- Amien, M. (2023). "Sejarah NLP Bahasa Indonesia." arXiv:2304.02746.
- Amien, M., Feng, C. & Huang, H. (2022). "Low-resource dataset creation for Indonesian local languages." arXiv:2206.07238.
- Amien, M., Chong, F. & Heyan, H. (2022). "Indonesian sub-word separator." arXiv:2207.00552.
- Ariesandy et al. (2020). "Synthetic augmentation for colloquial NMT." arXiv:2012.15178.
- Gunawan, G.F., Amien, M. & Palandi, J.F. (2017). "Chimera distributed computing framework." CAIPT 2017.
- Syafiandini et al. (2016). "Multimodal Deep Boltzmann Machines for gene expression feature selection." ACSI 2016.
- Syafiandini et al. (2016). "Cancer subtype identification using deep learning." IC3INA 2016.
- Fitriawan et al. (2016). "Deep Belief Networks for ligand-based virtual screening." WCSE 2016.
- Fitriawan et al. (2016). "Multi-label DBN for multi-target drug virtual screening." IC3INA 2016.
- Syafiandini et al. (2017). "RBM for gene expression in colorectal carcinoma." ICBB 2017.

RESEARCH GRANTS

- PDP DIKTI Grant (2013) — Indonesian Language Spam SMS Filtering Using Naive Bayes
- PDM DIKTI Grant (2011) — Network-Based Remote Surveillance Cameras

PROFESSIONAL AFFILIATIONS

- Member, Komunitas Dosen Indonesia (NUPN: 9907004473)

LANGUAGES

Bahasa Indonesia (native) | English (professional working proficiency)